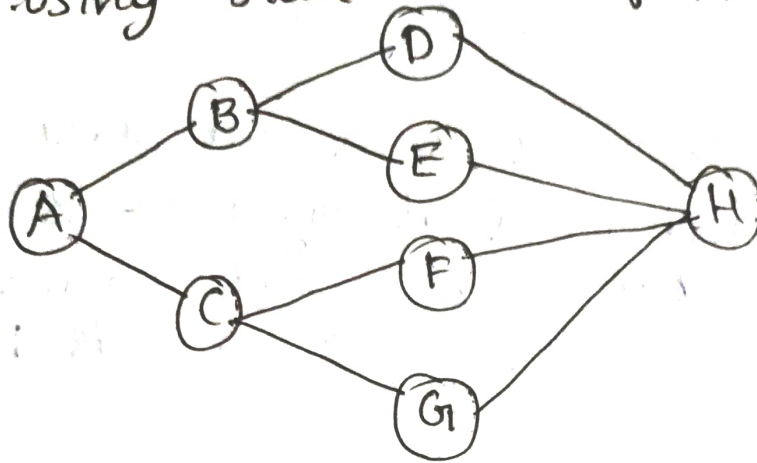


1) Explain the process of evaluating the postfix expression
expressing and evaluate the postfix expression

$$734+ - 245 / + 86 / 7 +$$

2) Explain BFS & DFS method for the following graph using start vertex of A.



3) Construct a binary tree ~~by with the~~ by considering the following tree traversal.

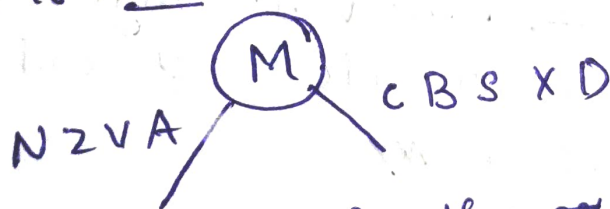
Inorder: N Z V A M C B S X D

Postorder: Z A V N C S D X B M

Answers

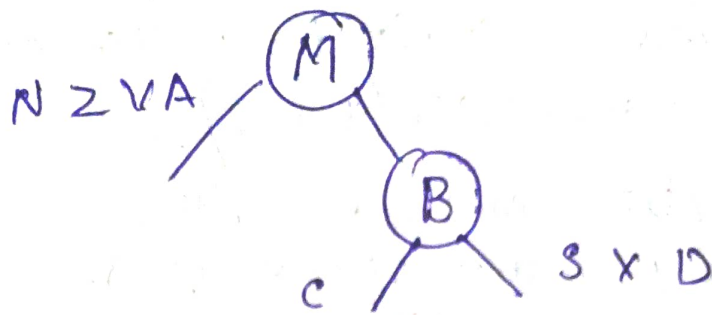
3A)

Step 1: Take the last node in the post order that is M. The M is the root node.

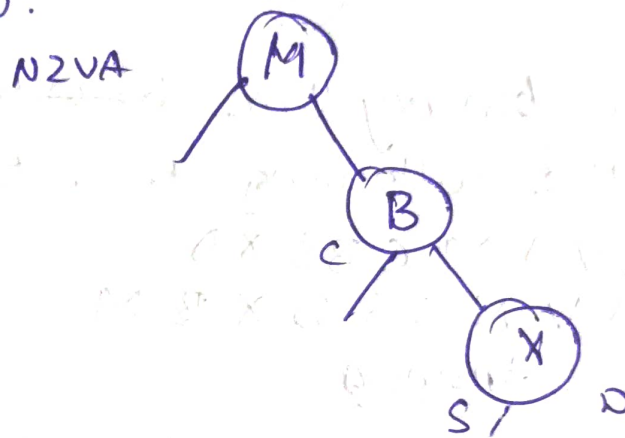


Now see the M in the ~~pre~~ Inorder. The left _{side} nodes of M are left subtree and right side node of M are right subtree.

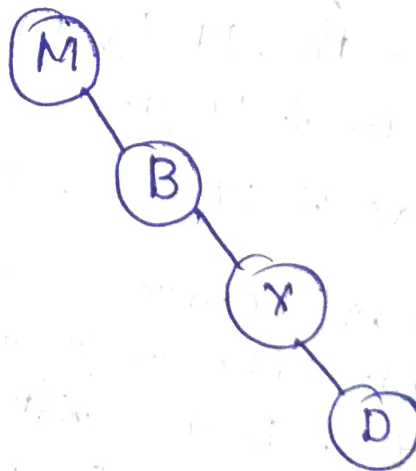
Step 2: The node before root node is B. It is the right child of root 'M'. and the nodes ~~left~~ left of B are left subtree of B and the nodes of right of B are right subtree of B.



Step 3: The node before B in post order is X. The X is ~~not~~ placed at right of B, and the nodes ~~left~~ of X is left subtree of X and the nodes right of X is right subtree of D.

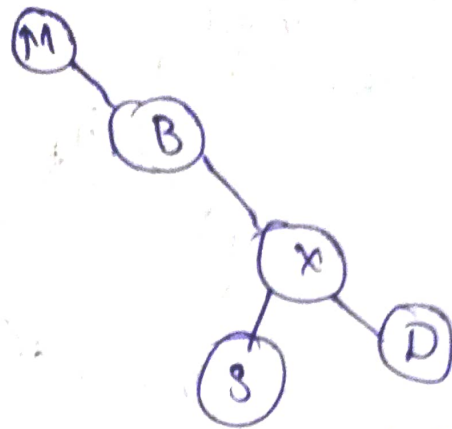


Step 4: The node before X in post order is D. The D is ~~the~~ placed at right of X. and there is no nodes left and right of D. ~~there is no~~



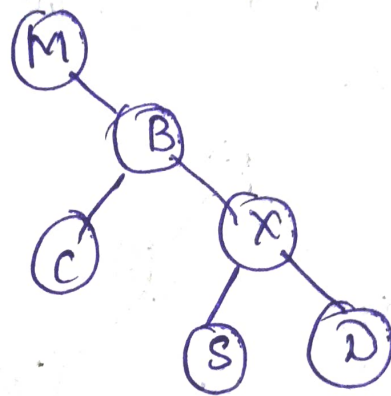
The D has no child, so left subtree of X is done.

step 5: The node before D in the post order is S.
S is the right node of X.



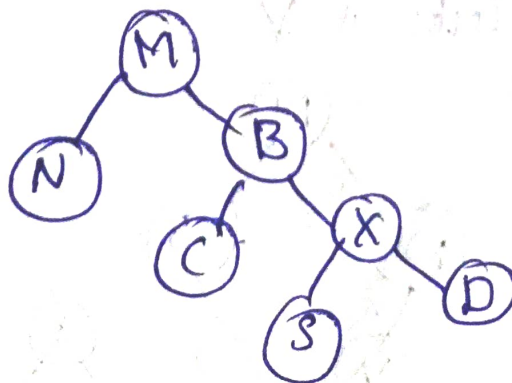
S has no right and left nodes, so the right subtree of X is done.

step 6: the node before S is C. C is the right node B by seeing the inorder positions.

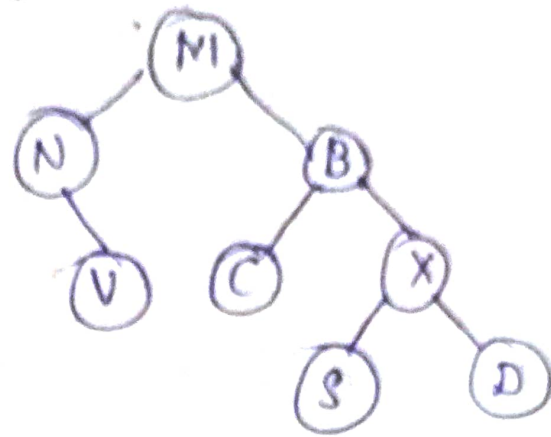


there is no right nodes ~~left~~ remain for root M. So the right subtree is completed.

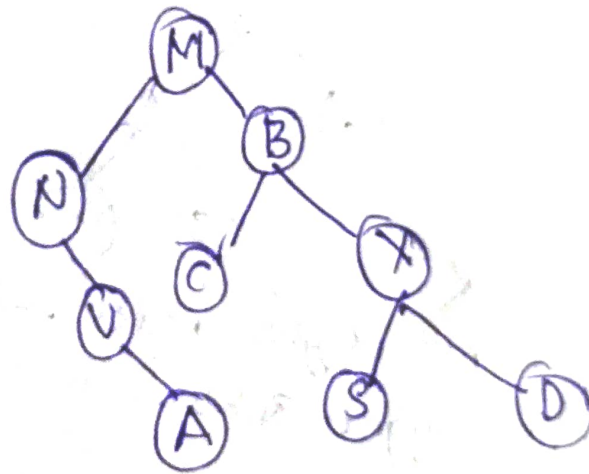
step 7) the node before C in the ~~in~~ preorder is N. N is the left node to the M based on inorder.



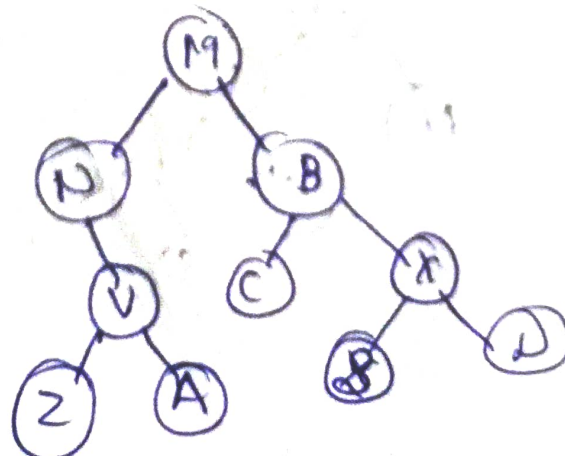
Step 8: The node before n in the post order is v .
 The v is ~~left~~^{right} of n in inorder. so v is the
~~left~~ right of n .



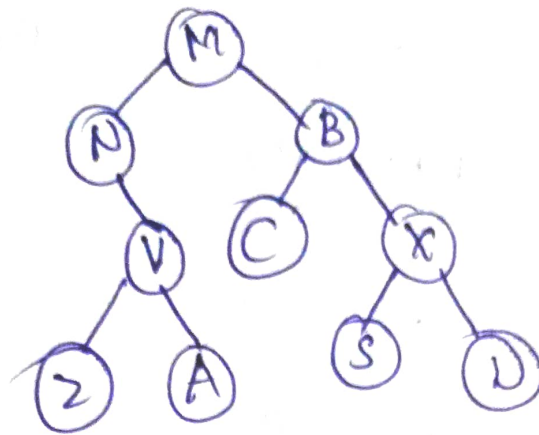
Step 9: The node before v in the post order is A .
 A is the right of v in inorder. so A is
 right of v .



Step 10: The node before A in post order is Z .
 Z is the left of v in inorder. so, Z is
 Placed left of v .



Step 11: There is no nodes to remain in the post order. so the final tree is



1A) Given Expression

$734+-245/+6/7+$

Algorithm of evaluate postfix Expression:

A Postfix expression is evaluated using a stack.

Rules:

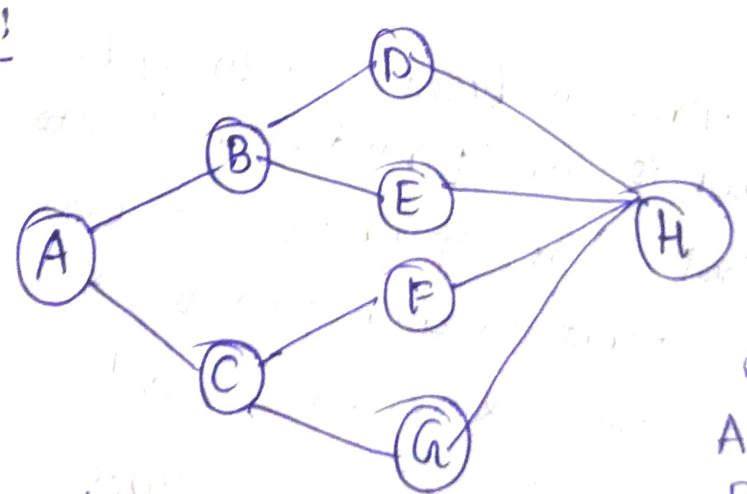
1. Scan the expression from left to right.
2. If the symbol is an Operand, Push it into the stack.
3. If the symbol is an operator, pop two Operands.
4. Perform the operation and push the result back.
5. Continue until the expression ends.

Reading character	Operation	Stack
7	Push	7
3	Push	7, 3
4	Push	7, 3, 4
+	POP(), POP() 4+3	7, 7
-	Pop, Pop 7-7	0

2	Push	0, 2
4	Push	0, 2, 4
5	Push	0, 2, 4, 5
/	Pop(), Pop() $4 \div 5 = 0.8$	0, 2, 0.8
+	Pop(), Pop() $2 + 0.8 = 2.8$	0, 2.8
*	Pop(), Pop() $0 \times 2.8 = 0$	0
6	Push	0, 6
/	$0 \div 6 = 0$	0
7	Push	0, 7
+	$0 + 7 = 7$	7

The result is 7

2A) DFS!



adj List

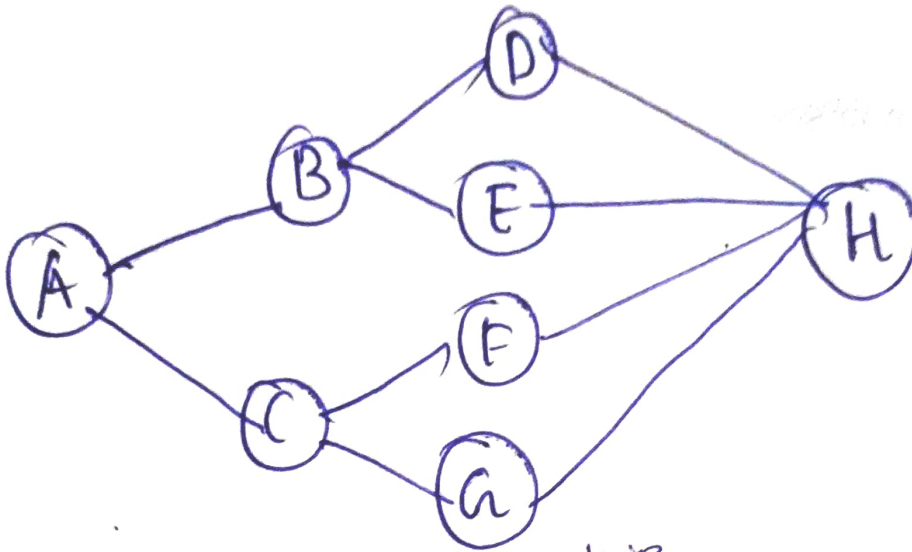
- A: B, C
- B: ~~G~~ D, E, A
- C: F, G, A
- D: H, B
- E: H, B
- F: H, C
- G: H, C
- H: D, E, F, G

DFS(G)
DFS(C)
DFS(F)
DFS(E)
DFS(H)
DFS(D)
DFS(B)
DFS(A)

Call stack

O/P, A, B, D, H, E, F, C, G

2) BFS



Adj List

A : B, C
B : D, E, A
C : F, G, A
D : H, B
E : H, B
F : H, C
G : H, C
H : G, F, E, D

Queue

~~A~~ ~~B~~ ~~C~~ ~~F~~ ~~G~~ ~~H~~ ~~H~~ ~~E~~ ~~D~~

skip

O/P : A B C F G H E D